

Chronic Kidney Disease Prediction Using Machine Learning Algorithms

K. Santhiya
Department of CSE
Manonmaniam Sundaranar University,
Abishekapatti, Tirunelveli 627012,
Tamil Nadu, India.
santhiya001kumar@gmail.com

D. Murugan
Department of CSE
Manonmaniam Sundaranar University,
Abishekapatti, Tirunelveli 627012,
Tamil Nadu, India.
dhanushkodim@yahoo.com

K. Navaz
Department of CSE
School of Computing
Vel Tech Rangarajan Dr. Sagunthala
R&D Institute of Science and
Technology, Avadi, Chennai 600062,
Tamil Nadu, India.
navazit@gmail.com

Abstract— Chronic Kidney Disease (CKD) stands as a pervasive and critical renal condition, posing a substantial global health challenge. Each kidney comprises nephrons, the essential filtration units that help to filter the blood and extract the waste products. Individuals with type 2 diabetes or high blood pressure face an elevated risk of renal failure. Predicting renal failure at an early stage remains challenging due to the absence of specific symptoms, leading to severe long-term consequences, including mortality. This paper aims to comprehensively analyze and assess the efficacy of predictive models for CKD, using various Machine Learning Classification Algorithms. The classifiers such as K-Nearest Neighbor (KNN), Decision Tree (DT), Random Forest (RF), Naïve Bayes (NB), and Gradient Boosting (GB). Out of all the classifiers Random Forest and Gradient Boosting produced the best accuracy result compared to other classifiers. This research extends the investigation by providing a nuanced analysis of the significance of machine learning techniques in predicting CKD, emphasizing their role in early detection and prognosis. This research includes a detailed comparison of accuracy scores and, a confusion matrix to identify correct and incorrect features, thereby facilitating a comprehensive examination and analysis of machine learning algorithms. The evaluation utilizes accuracy, precision, recall, and f1-score to detect the optimal model.

Keywords—CKD, Machine Learning, Classification Algorithms

I. INTRODUCTION

The kidney, a pivotal organ situated proximately to the lower rib cage, plays a crucial role in maintaining the body's electrolyte balance. It orchestrates the equilibrium of saltwater and various ions, encompassing essential vitamins and minerals such as calcium, phosphorus, magnesium, potassium, chlorine, and diverse acids. Within each kidney reside approximately one million nephrons, consisting of glomeruli and tubules that serve as indispensable filters. The glomerulus filters blood, while the tubule reinstates vital nutrients and expels waste products. Chronic kidney failure, an insidious progression marked by the constant elevation of creatinine and blood urea, unfolds surreptitiously without manifesting signs or symptoms until irreparable damage occurs. This deterioration culminates in the imperative need for interventions like dialysis or kidney transplants for survival. Early analysis proves pivotal in averting the onset of kidney failure. Evaluating kidney function and predicting disease stages hinge on observing the Glomerular Filtration Rate (GFR), computed based on age, gender, and blood

creatinine levels. In India alone, a staggering one million new instances of CKD annually.

Stages of CKD:

Stage 1 - Kidney Damage with Normal GFR

In this initial phase, kidney damage subtly unfolds without overt symptoms. Detectable indicators include abnormal protein or blood levels in urine tests. The emphasis lies on instigating lifestyle changes and managing underlying conditions.

Stage 2 - Mild Reduction in GFR

GFR experiences a mild decline, signaling early impairment. While symptoms may remain minimal or absent, heightened vigilance becomes imperative. Dietary adjustments, blood pressure control, and regular monitoring are pivotal in decelerating progression.

Stage 3 - Moderate Reduction in GFR

A moderate decline in kidney function ushers in more noticeable symptoms. Fatigue, fluid retention, and alterations in urination patterns become apparent. A comprehensive approach involves dietary restrictions on protein, sodium, phosphorus, and medication.

Stage 4 - Severe Reduction in GFR

This stage necessitates heightened medical attention. Conditions such as hypertension, anemia, and bone disorders may emerge. A multidisciplinary approach with nephrologists, dietitians, and specialists becomes vital. Preparations for renal replacement therapy, like dialysis, intensify.

Stage 5 - End-Stage Kidney Disease (ESKD)

ESKD demands immediate intervention as symptoms intensify, affecting cardiovascular, hematologic, and musculoskeletal systems. Dialysis becomes a lifeline, and the pursuit of kidney transplantation becomes paramount for an improved quality of life.

Machine Learning in CKD Prediction:

This research harnesses the power of Machine Learning (ML) models to predict CKD. These models, including K-Nearest Neighbor (KNN), Decision Tree (DT), Random Forest (RF), Naïve Bayes (NB), and Gradient Boosting (GB), demonstrate their performance in analyzing historical records of CKD-affected patients. This paper also examines the most accurate strategy for predicting CKD. The result of this study shows that ML models are not only capable of carrying out this crucial clinical task but also have the potential to enhance personalized medicine. The rest of the paper is organized as follows Section 2 provides an in-depth exploration of existing research in the field of

CKD. In section 3, the methodology employed in this study is elucidated. Section 4 enlightens Results and discussion, and section 5 succinctly summarizes the entirety of the Research work.

II. RELATED WORK

The emergence of Machine Learning and improvements in computing capabilities has opened up a number of new research options in the field of healthcare. Various researchers have proposed Machine Learning algorithms to enhance the accuracy of CKD disease prediction. The research conducted by Harsh Vardhan Singh [1], the primary focus was on evaluating the feasibility of utilizing Machine Learning algorithms to predict Chronic Kidney Disease with a historical CKD dataset. The author explored various classifiers, including Support Vector Machine, Naive Bayes, Decision Tree, Logistic Regression, K-Nearest Neighbor, and Random Forest, all aimed at achieving precise CKD failure prediction. The in-depth analysis included a meticulous examination of each classifier's performance, considering metrics such as accuracy, precision, recall, and F1-score. Among the Machine Learning algorithms scrutinized, the study found that Support Vector Machine (SVM) and Decision Tree exhibited superior accuracy and robust performance across diverse evaluation criteria.

Akalya Devi C et al [2] conducted a comprehensive literature review. This involved a meticulous exploration of existing studies dedicated to CKD prediction methodologies. Their examination likely delved into the methodologies, datasets, and outcomes of prior research, providing insights into the landscape of CKD prediction. By analyzing the research, the authors could have identified trends, addressed gaps, and critically evaluated the effectiveness of different data mining algorithms for CKD prediction.

Gazi Mohammed Ifraz et al [3] undertook a study on CKD, utilizing a dataset with 400 records of kidney patients. Their research involved applying Decision Tree, K-Nearest Neighbor, and Logistic Regression algorithms for CKD prediction. The results highlighted the effectiveness of Decision Tree and Logistic Regression in achieving high accuracy during CKD classification. Through cross-validation measures, they concluded that Logistic Regression outperformed other models in predicting kidney disease.

qiong bai et al [4] along with their team conducted an extensive study involving four distinct machine learning classification algorithms lr nb dt rf and knn their investigation aimed to evaluate the effectiveness of these algorithms in predicting chronic kidney disease, additionally, they employed roc curves to analyze the performance of random forest and the kidney failure risk equation model. The research findings underscore a critical insight the prognosis of CKD relies significantly on the early accessibility of specific features this thorough exploration of various machine learning algorithms and their corresponding performance metrics provides a unique and valuable contribution to the field of CKD prediction emphasizing the pivotal role of early detection of ckd.

Vineeta Gulati et al[5] presented a robust model to identify chronic kidney disease ckd in existing patient records the research involved an in-depth comparison of various machine learning and deep learning algorithms to determine the optimal model for ckd prediction among the plethora of algorithms explored including multi-layer perceptron mlp support vector machine svm and recurrent neural network, the research findings highlighted their superior accuracy in predicting ckd this nuanced exploration provided comprehensive insights into the strengths of different algorithms paving the way for more effective strategies in ckd diagnosis the significance of this study lies in its meticulous approach to evaluating and comparing algorithms contributing to the growing body of knowledge in leveraging advanced technologies for accurate ckd prediction from patient records.

Elias Dritsas et al [6] extensively explored a wide range of Machine Learning Algorithms to effectively tackle the non-uniform distribution of instances in predicting Chronic Kidney Disease. Their research was dedicated to devising robust solutions to overcome challenges associated with data imbalances. In summary, this study makes a significant contribution to refining CKD prediction models by showing how Machine Learning Algorithms adapt to intricate data patterns, ultimately enhancing the overall accuracy of predictions.

Parul Sinha et al [7] conducted a comprehensive evaluation of various Machine Learning and Data Mining algorithms for predicting Chronic Kidney Disease (CKD). The study specifically focused on comparing the performance metrics of K-Nearest Neighbor (KNN) and Support Vector Machine (SVM). KNN demonstrated superior accuracy. This detailed examination provides valuable insights into the specific strengths of KNN in the context of CKD prediction, contributing to a nuanced understanding of suitable algorithm for accurate and reliable CKD pre its potential applications in healthcare scenarios.

Khaled Mohamad Almustafa [9] conducted a comprehensive study employing a range of Machine Learning (ML) algorithms for the classification of Chronic Kidney Disease (CKD) datasets. The algorithms utilized in this research included Random Tree, K-nearest neighbor, J48, Stochastic Gradient Descent (SGD), Decision Tree (DT), and Naive Bayes (NB) Classification Algorithms. Through a meticulous analysis of the results, it was observed that J48 and Decision Tree Classifier exhibited superior accuracy compared to the other algorithms employed in the classification process.

In the exploration conducted by Garima Shukla et al[10], a set of machine learning techniques, including K-nearest neighbors (KNN), Decision Tree, and Artificial Neural Network (ANN), were employed to predict Chronic Kidney Disease (CKD). Their research focused on identifying crucial attributes pivotal for the early prediction of CKD. Among the three algorithms in their work, Decision Tree emerged with noteworthy results, exhibiting an impressive 98.60% accuracy. This emphasizes the efficacy of Decision Tree in the context of CKD prediction and underscores its potential for high-precision healthcare applications.

In the study conducted by Md Mahedi et al [10], a comprehensive exploration of multiple machine learning (ML) algorithms for predicting Chronic Kidney Disease was

undertaken. Notably, the researchers addressed challenges like missing data by implementing K-nearest neighbors (KNN) imputation techniques. To overcome imbalanced data concerns, the Synthetic Minority Over-sampling Technique (SMOTE) was employed. Feature selection played a crucial role, with the researchers utilizing correlation-based feature selection and Principal Component Analysis (PCA) to refine the dataset. The selected features were then fed into various ML algorithms, including KNN, Support Vector Machine (SVM), Gaussian Naive Bayes, Decision Tree, Logistic Regression, and ensemble learning techniques. Remarkably, the ensemble learning algorithms demonstrated superiority over other techniques, showcasing the effectiveness of a combined approach in CKD prediction. This research contributes valuable insights into handling data challenges and optimizing ML models for enhanced predictive accuracy in the context of CKD.

III. METHODOLOGY

To predict CKD, the process begins with a Kaggle dataset consisting of 400 patient records with both numerical and categorical features[11]. The initial step involves preprocessing, where missing values are addressed using Random Sampling and mean/mod sampling to ensure the dataset's integrity. Categorical attributes are then transformed into numerical values through Label Encoding. The dataset is split into 70% for training and 30% for testing, facilitating the training of machine learning models such as Random Forest, Support Vector Machine, and K-Nearest Neighbor. Model performance is assessed using accuracy, precision, recall, and F1-score to ensure effective CKD prediction. Figure 1 shows the framework of the research.

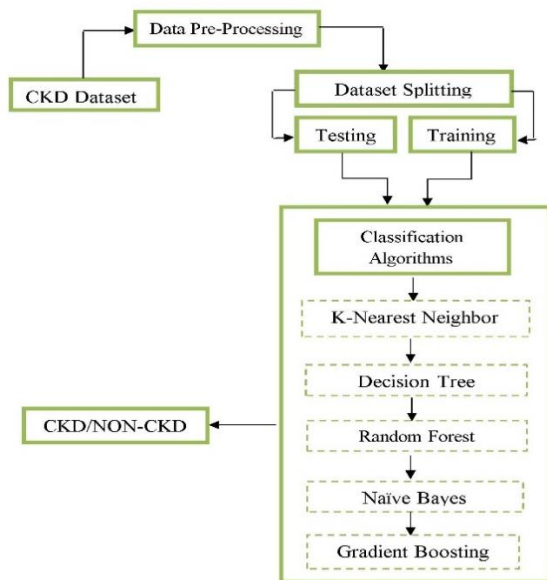


Figure. 1 Framework of the research work

A) K- Nearest Neighbor (K-NN):

The K-Nearest Neighbor (KNN) algorithm is the simplest and lazy learning algorithm, serving as a

cornerstone in machine learning. In the realm of CKD prognosis, KNN leverages all available patient attributes in the dataset, including factors like blood pressure, sugar levels, serum creatinine, and albumin etc. The decision-making process relies on a majority vote from the nearest neighbors, with the parameter 'K' representing the chosen number of neighbors [12]. Below are the sequential steps of the algorithm:

Step 1: Initialization:

- Begin the process.

Step 2: Data Import:

- Integrate the training and testing datasets.

Step 3: Neighbor Selection:

- Specify the number of neighbors, 'K.'

Step 4: Euclidean Distance Calculation:

- For each test data point:

Compute the Euclidean distance to every training data point.

- Store and sort these distances.

Step 5: Categorization of Data Points:

- Examine the distribution among the K nearest neighbors.
- Assign new data points to the category with the highest neighbor count.

Step 6: Conclusion:

- End the process.

B) Decision Tree (DT):

Decision Tree Algorithm emerges as a multifaceted and adaptive solution for CKD. Decision Trees meticulously analyze an array of patient attributes. These encompass crucial factors such as blood pressure, serum creatinine levels, age, and an expansive spectrum of health indicators[13]. The following outlines the steps associated with decision tree algorithm

Step 1: Initialization

- Begin the process of constructing an effective decision tree.

Step 2: Data Integration

- Integrate training and testing datasets, laying the foundation for comprehensive analysis.

Step 3: Attribute Evaluation:

- Assess each attribute meticulously, calculating both information gain and entropy to ensure a thorough understanding.

Step 4: Decision Node Formation:

- Establish a decision node by selecting the attribute that maximizes information gain, a critical step for robust decision-making.

Step 5: Child Node Formation:

- Elaborate on child nodes iteratively, evaluating information gain and entropy for remaining attributes, fostering a nuanced and well-balanced tree structure.

Step 6: Precision-Focused Pruning:

- Implement precision-focused pruning to refine the decision tree, optimizing its architecture for enhanced predictive accuracy.

Step 7: Recursive Iteration:

- Recursively refine the decision tree, achieving optimal attribute coverage while maintaining a delicate balance between complexity and interpretability.

Step 8: Conclusion:

- End the Process

C) Random Forest (RF):

In medical diagnostics, Random Forest plays a crucial role in predicting cases of CKD and non-CKD. The ensemble strategy ensures robust predictions, establishing RF as a valuable asset in CKD classifications[14]. Below are the steps involved in the Random Forest algorithm.

Step 1: Initialization

- Start the prediction process.

Step 2: Import Data

- Ingest the required training and testing datasets.

Step 3: Decision Tree Predictions:

- Utilize randomly created decision trees to predict and store outcomes for the given test data.

Step 4: Aggregate Votes:

- Calculate total votes for each class based on individual decision trees' predictions.

Step 5: Majority Class Declaration

- Identify the class with the highest vote count as the final predicted outcome class.

Step 6: Final Prediction:

- Assign the majority class as the final predicted outcome.

Step 6: Conclusion:

- End the prediction process

D) Naive Bayes (NB):

Naive Bayes algorithm works in the principles of Bayes' theorem, Naive Bayes operates by evaluating the likelihood of each feature occurring within every class and establishing the prior probability of each class. The following steps outline the key process involved in the implementation of the Naive Bayes algorithm.

Step 1: Initialization

- The process begins by initializing the Naive Bayes algorithm for classification.

Step 2: Import Data

- Import the required training and testing datasets

Step 3: Parameter Calculation

- Compute the mean and standard deviation of the predictor variables within each class.

Step 4: Probability Computation

- Utilize the Gaussian density equation to calculate the probability for predictor variable ($f_1, f_2, f_3, \dots, f_n$) in each class. Repeat this process until all probabilities are determined.

Step 5: Likelihood Determination

- Calculate the likelihood for each class based on the computed probabilities.

Step 6: Conclusion

- End the process

E) Gradient Boosting (GB):

Its transformative capabilities are particularly noteworthy in predictive modeling, providing a level of

precision that goes beyond conventional methods. The algorithm signifies vital health-related factors, encompassing attributes like creatinine levels and other pertinent parameters. This strategic approach enhances predictive accuracy and empowers healthcare professionals with an understanding of early detection and precise classification of chronic kidney disease cases. The step-by-step process of the Gradient Boosting Algorithm is outlined below.

Step 1: Initialization

- Start the prediction process

Step 2: Import Data

- Ingest the relevant training and testing datasets.

Step 3: Calculate Average

- Compute the average of the target label to establish a baseline.

Step 4: Calculate Residuals

- Determine residuals by finding the difference between the actual and predicted values.

Step 5: Decision Tree Construction

- Build a decision tree focusing on the identified residuals.

Step 6: Predict with Trees

- Utilize the ensemble of trees to predict the target label.

Step 7: Compute New Residuals

- Assess and calculate new residuals based on the latest predictions.

Step 8: Iterative Process

- Iteratively refine predictions by repeating steps 4 to 7 until reaching the specified iteration count (controlled by a hyper parameter).

Step 9: Final Prediction

- Once the algorithm is adequately trained, leverage the entire ensemble of trees to formulate a comprehensive final prediction for the target variable.

Step 10: Conclusion

- Conclude the algorithmic execution.

F) Performance Analysis

The primary objective of performance evaluation is to ascertain how well a model generalizes to unknown or out-of-sample data [15]. To gauge the performance of machine learning models, a diverse set of metrics such as precision, recall, accuracy, and F-Measure are considered, each provides a unique perspective on the model's real-world efficacy. Furthermore, delving into performance metrics like True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) provides an understanding of the model's prediction.

- True Positive (TP - CKD): It indicates the model's proficiency in correctly identifying and classifying individuals with CKD.
- True Negative (TN - Non-CKD): This underscores the model's accuracy in correctly recognizing and excluding individuals without CKD.
- False Positive (FP - Non-CKD Misclassification): This metric reveals, where the model may generate a positive prediction erroneously, leading to a misclassification of non-CKD cases.

- False Negative (FN - CKD Misclassification): This highlights, where the model fails to identify CKD cases, resulting in misclassification of individuals with the condition.

The formula for Accuracy, Precision, Recall, and F1-Score are discussed below.

Accuracy:

Accuracy is the measure of the ratio of correct predictions over the total number of instances [16].

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

Precision:

From the total predicted values in the actual class, precisely measure the true values that were correctly predicted. Precision measures a classifier's capacity to avoid misclassifying a negative example as positive.

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

Recall:

The proportion of positive values that are correctly classified is called the recall rate. The percentage of true positives that are correctly classified by the recall.

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

F1-Score:

F-measure is the harmonic mean of the precision and recall and summarizes the predictive accuracy of a model.

$$F1 = 2 \cdot \frac{precision \cdot recall}{precision + recall} \quad (4)$$

IV. RESULTS AND DISCUSSION

In this study, the CKD prediction has been carried out using Machine Learning techniques. Several related studies have been conducted to predict CKD using machine learning and other techniques globally. However, they have limited work and they have mainly focused on binary class classification. The dataset has been taken from the Kaggle repository it consists of 26 attributes with numerical and nominal values along with the corresponding class for each instance. With 400 instances featuring missing values and categorical data, a two-step pre-processing strategy was implemented.

The dataset comprises 400 instances with missing values and categorical data. In this research work, the pre-processing has been done in two steps. To eliminate undesirable noise from the dataset the preprocessing must be done beforehand. In the first step, the missing values were filled using the Random sampling method. In the second step, the categorical data was changed into numerical value using the Label Encoding method, and the data is split in the ratio of 70:30. Training data is a subset of the original data

that is employed to build the model however testing data is used to evaluate the model's accuracy. Finally, Machine Learning algorithms such as KNN, DT, RF, NB, and GB algorithms are used for the prediction of CKD.

Moreover, Table 1 illustrates all the classifier's performance. KNN faced challenges rooted in its lazy algorithmic nature, translating into unsatisfactory results. Computational burdens, particularly in classifying test samples, coupled with memory overhead, underscored the drawbacks of this approach. [17]. Decision Tree, mirroring human decision-making processes, emerged as an interpretable and efficient choice. With a notable accuracy of 94% in CKD prediction, it shows its efficacy in both discrete and continuous input scenarios [18]. The ensemble learning approach of Random Forest building multiple trees on data subsets yielded a remarkable accuracy of 98%. This not only addressed Decision Tree's overfitting concerns but also presented an exemplary prediction outcome [19]. Naive Bayes theorem shows a computational capability. In the case of CKD prediction, it delivered a commendable accuracy score of 96.5%. Gradient Boosting utilizes gradient descent to pinpoint the challenges in the learner's prediction used in the previous tree. The previous error is highlighted and combines one weak learner with the next learner so the error is reduced significantly [20] and it produced an accuracy of 97%.

Table 1 Overall performances of the ML algorithms

CLASSIFIERS	ACCURACY (%)
KNN	69%
DT	94%
RF	98%
NB	96%
GB	97%

Table 1 provides a overall performance metrics of the classification, the following below elaborates why each algorithm exhibits varying levels of accuracy in classifying Chronic Kidney Disease (CKD).

1) Random Forest Dominance (Accuracy: 98%)

The ensemble approach of Random Forest involves building multiple trees on data subsets, mitigating Decision Tree's overfitting. This concerted effort results in accurate prediction.

Reason for High Accuracy: The combination of multiple trees, each contributing its prediction, enhances accuracy by minimizing overfitting and capturing diverse patterns in the CKD dataset.

2) Gradient Boosting Excellence (Accuracy: 97%)

Gradient Boosting employs an iterative refinement process, addressing challenges highlighted by its predecessors. It sequentially combines weak learners to significantly reduce errors and enhance accuracy.

Reason for High Accuracy: The iterative refinement process allows Gradient Boosting to adapt and improve with each iteration, refining its predictions and achieving high accuracy.

3) Naive Bayes and Decision Tree Competence (Accuracy: 96% and 94%, respectively)

Naive Bayes Strengths: Naive Bayes, based on the Bayes theorem, works efficiently and accurately in CKD prediction, using probabilities for classification.

Decision Tree Strengths: Decision Tree mirrors human decision-making and excels in transparently understanding and predicting outcomes.

Reason for High Accuracy: Both algorithms capitalize on their inherent strengths, utilizing probability distributions (Naive Bayes) and transparent decision-making (Decision Tree) for accurate CKD prediction.

4) KNN Challenges Persist (Accuracy: 69%):

KNN faces computational challenges, especially with large datasets. The lazy algorithmic nature and sensitivity to imbalanced sample sizes hinder its efficiency.

Reason for Lower Accuracy: Computational overhead, memory constraints, and sensitivity to imbalanced data impact KNN's ability to classify CKD instances accurately, resulting in lower accuracy.

V. CONCLUSION

Chronic kidney disease is a condition characterized by progressive loss of kidney function over time. Most patients have no symptoms, making it a silent disease. This paper presents a comparative analysis of machine learning techniques using a historical patient dataset with various features related to chronic kidney disease. Machine Learning algorithms such as K-Nearest Neighbor, Decision Tree, Random Forest, Naive Bayes, and Gradient Boosting are used for classification. The performance of Random Forest was found to be 98% and the Gradient Boosting method was found to be 97% accurate compared to KNN (69%), Naive Bayes (93%), and Decision Tree (94%). It was found that the Random Forest algorithm and Gradient Boosting algorithm work better than other machine learning classification algorithms. The following points are suggested for future works

- The work can be expanded further while taking various factors like junk food intake, intake of clean water, and various environmental factors.
- The accuracy of CKD prediction can be further improved by using Deep Learning algorithms like Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Multi-Layer Perceptron (MLP), or by using other ensemble learning techniques.

Conflicts of Interest

The authors declare that they have no known competing financial interest or personal relationships that would have appeared to influence the work reported in this paper.

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